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Probabilistic Combinatorics SS24

Sheet 7

Due: 3. June at 19:00

(1) Show the Principle of Mutual Independence (Lemma 10.1) from the lecture.

(2) Hash Functions
Let X and Y be finite sets with $|X| = n$ and $|Y| = m$. Furthermore let $\mathcal{F}$ be a family of N maps $f : X \to Y$ and $1 \leq w \leq n$. Show: If

$$N > \frac{\log(e \cdot (\binom{n}{w} - \binom{n-w}{w}))}{\log(m^w) - \log(m^w - w!\binom{m}{w})}$$

then there is a family $\mathcal{F}$ with $|\mathcal{F}| = N$, such that for every w-element subset C of X, there is a $f \in \mathcal{F}$ such that $f_{|C}$ is injective.

A familiy $\mathcal{F}$ of maps from a n-element to a m-element set, satisfying the conditions above, is called a (n, m, w)-perfect family of hash functions. These play an important role in computer science when storing large amounts of data.

(3) Lovasz-Local Lemma; k-Conjunctive Normal Forms
A boolean variable is a variable X, that only takes values **TRUE** or **FALSE**. A literal is a boolean variable or its negation. A k-clause is a disjunction of k literals of different variables. A formula in k-conjunctive normal form is a conjunction of k-clauses.

Show: If F is a formula in k-conjunctive normal form, such that every variable appears in at most $\frac{1}{k} \cdot \frac{2^k}{e} - 1$ clauses, then the appearing boolean variables can take values such that the formula evaluates to **TRUE**.

(4) Let $(S_n, \mathbb{P})$ be the probability space of permutations with the uniform probability. For $i = 1, ..., n-1$ let A_i denote the event of all $\sigma \in S_n$ with $\sigma(i) > \sigma(i+1)$.

(i) Determine which of the A_i are independent of each other and which are not.

(ii) Decide, wheather there is an independence graph and a choice of $x_i \in [0, 1)$, $i = 1, ..., n-1$, such that using the Lovasz-Local-Lemma we can proof

$$\mathbb{P}(\bigcap_{i=1}^{n-1}(S_n \setminus A_i)) > 0.$$

If not: Is the event empty?